
Homework_5: CFG & CNF

Question_1:

Find CFG for $L = \{a^n b^n c^m d^m : n, m \geq 0\}$

$S \rightarrow AB$

$A \rightarrow aAb \mid \epsilon$

$B \rightarrow cBd \mid \epsilon$

Derivation test: aabbcd

$S \rightarrow AB \rightarrow aAbB \rightarrow aaAbbB \rightarrow aa\epsilon b b c B d \rightarrow a a b b c \epsilon d \rightarrow a a b b c d$

Question_2:

Convert the following CFG to CNF

$S \rightarrow AaB \mid b$

$A \rightarrow S \mid \epsilon \mid AB$

$B \rightarrow bbb \mid ASA$

Step1 : eliminate start symbol from the RHS

-Add new product Rule

$$S_0 \rightarrow S$$

$$S \rightarrow AaB \mid b$$

$$A \rightarrow S \mid \varepsilon \mid AB$$

$$B \rightarrow bbb \mid ASA$$

Step2 : eliminate null production

$$S_0 \rightarrow S$$

$$S \rightarrow AaB \mid b$$

$$A \rightarrow S \mid \varepsilon \mid AB$$

$$B \rightarrow bbb \mid ASA$$

$$S_0 \rightarrow S$$

$$S \rightarrow AaB \mid aB \mid b$$

$$A \rightarrow S \mid AB \mid B$$

$$B \rightarrow bbb \mid ASA \mid SA \mid AS \mid S$$

Step3 : eliminate unit production (Remove $S \rightarrow AaB \mid aB \mid b$)

$$S_0 \rightarrow S$$

$$S \rightarrow AaB \mid aB \mid b$$

$$A \rightarrow S \mid AB \mid B$$

$$B \rightarrow bbb \mid ASA \mid SA \mid AS \mid S$$

$$S_0 \rightarrow AaB \mid aB \mid b$$

$$S \rightarrow AaB \mid aB \mid b$$

$$A \rightarrow AaB \mid aB \mid b \mid AB \mid B$$

$$B \rightarrow bbb \mid ASA \mid SA \mid AS \mid AaB \mid aB \mid b$$

Step3 cont. : eliminate unit production (Remove $B \rightarrow bbb \mid ASA \mid SA \mid AS \mid AaB \mid aB \mid b$)

$S_0 \rightarrow AaB \mid aB \mid b$

$S \rightarrow AaB \mid aB \mid b$

$A \rightarrow AaB \mid aB \mid b \mid AB \mid B$

$B \rightarrow bbb \mid ASA \mid SA \mid AS \mid AaB \mid aB \mid b$

$S_0 \rightarrow AaB \mid aB \mid b$

$S \rightarrow AaB \mid aB \mid b$

$A \rightarrow AaB \mid aB \mid b \mid AB \mid bbb \mid ASA \mid SA \mid AS$

$B \rightarrow bbb \mid ASA \mid SA \mid AS \mid AaB \mid aB \mid b$

Step4 : replace terminals in mixed production ($X \rightarrow a, Y \rightarrow b$)

$S_0 \rightarrow AaB \mid aB \mid b$

$S \rightarrow AaB \mid aB \mid b$

$A \rightarrow AaB \mid aB \mid b \mid AB \mid bbb \mid ASA \mid SA \mid AS$

$B \rightarrow bbb \mid ASA \mid SA \mid AS \mid AaB \mid aB \mid b$

$S_0 \rightarrow AXB \mid XB \mid b$

$S \rightarrow AXB \mid XB \mid b$

$A \rightarrow AXB \mid XB \mid b \mid AB \mid YYY \mid ASA \mid SA \mid AS$

$B \rightarrow YYY \mid ASA \mid SA \mid AS \mid AXB \mid XB \mid b$

$X \rightarrow a$

$Y \rightarrow b$

Step5 : eliminate RHS with more than two non-terminals

$$S_0 \rightarrow AXB \mid XB \mid b$$

$$S_0 \rightarrow ZB \mid XB \mid b$$

$$S \rightarrow AXB \mid XB \mid b$$

$$S \rightarrow ZB \mid XB \mid b$$

$$A \rightarrow AXB \mid XB \mid b \mid AB \mid YYY \mid ASA \mid SA \mid AS$$

$$A \rightarrow ZB \mid XB \mid b \mid AB \mid TY \mid RA \mid SA \mid AS$$

$$B \rightarrow YYY \mid ASA \mid SA \mid AS \mid AXB \mid XB \mid b$$

$$B \rightarrow TY \mid RA \mid SA \mid AS \mid ZB \mid XB \mid b$$

$$X \rightarrow a$$

$$X \rightarrow a$$

$$Y \rightarrow b$$

$$Y \rightarrow b$$

$$Z \rightarrow AX$$

$$R \rightarrow AS$$

$$T \rightarrow YY$$

Final Productions:

$S_0 \rightarrow ZB \mid XB \mid b$

$S \rightarrow ZB \mid XB \mid b$

$A \rightarrow ZB \mid XB \mid b \mid AB \mid TY \mid RA \mid SA \mid AS$

$B \rightarrow TY \mid RA \mid SA \mid AS \mid ZB \mid XB \mid b$

$X \rightarrow a$

$Y \rightarrow b$

$Z \rightarrow AX$

$R \rightarrow AS$

$T \rightarrow YY$